

Uncertainty-aware haplotype phasing of genetic and epigenetic variation from long reads

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Abstract

Long reads link genetic variants and native methylation signals, but integrating these observations into reliable haplotypes remains challenging in repetitive regions. Here we present LongPhase 2, which co-phases SNVs, indels, structural variants and allele-specific methylation in one graph, uses quality- and context-aware weighting, and scores every variant's phasing uncertainty. A graph neural network then unphases, rather than flips, the variants it predicts to be misphased. On HG002 nanopore data (10–60×), against a telomere-to-telomere truth set, LongPhase 2 makes 2.8- to 4.8-fold fewer switch errors than WhatsHap and HapCUT2 (1.1- to 1.7-fold under GIAB v4.2.1) and runs six- to eightfold faster at 60×; co-phasing lengthens haplotype blocks by 40%. On PacBio HiFi data, without retraining the network, LongPhase 2 makes 2.5- to 3.5-fold fewer switch errors than WhatsHap. LongPhase 2 is a drop-in replacement for the LongPhase versions in Clair3, ClairS, Severus and EPI2ME.

Introduction

Haplotype phasing, the assignment of heterozygous variants to their parental chromosomes, is essential for interpreting the diploid human genome. Whether two variants act in *cis* or in *trans* decides compound heterozygosity in recessive disease¹, and allele-specific methylation on a phased haplotype reveals its parent of origin². Long-read sequencing has made read-based phasing practical at clinical and population scale. Individual reads span tens to hundreds of kilobases, link distant heterozygous sites directly and carry native base-modification signals; a single library thus yields haplotype blocks orders of magnitude longer than short reads allow^{3,4}. Read-based phasing also needs nothing beyond the sample's own reads: no sequencing of the parents, no population reference panel of the kind statistical phasing depends on⁵, and none of the additional Hi-C or Strand-seq libraries otherwise used to phase whole chromosomes⁶.

Read-based phasers optimize agreement between reads and candidate haplotypes through weighted minimum error correction or probabilistic models, as in WhatsHap⁷, HapCUT2⁸ and Margin⁹. These methods centre on single-nucleotide variants (SNVs), although WhatsHap, by re-aligning each read to both alleles, also co-phases small and large indels with them¹⁰. We previously introduced LongPhase¹¹, which recasts phasing as a weighted allele graph, resolves

haplotypes at chromosome scale and co-phases SNVs with structural variants (SVs)^{12,13}. WhatsHap and LongPhase now sit beneath much of the long-read analysis stack. Clair3 phases with WhatsHap by default and with LongPhase as an option, and the somatic callers ClairS and ClairS-TO use LongPhase by default^{14–17}; the EPI2ME wf-human-variation workflow phases every nanopore genome with LongPhase¹⁸; and the SV callers Sniffles2 and Severus take as input the haplotagged alignments that either phaser writes^{12,19}. An error made by the phaser is therefore inherited, without any warning, by every analysis downstream.

Three challenges remain in read-based phasing. The first is integrating 5-methylcytosine (5mC), which long reads report alongside the sequence, into haplotype inference. Where the two haplotypes differ in methylation, as at imprinted loci and at sites of sequence-dependent allele-specific methylation, a read’s CpG states tie it to one haplotype even where it spans no heterozygous variant, which is where sequence-based blocks end^{20,21}. Several methylation-aware tools use that signal once a sequence phase exists. NanoMethPhase assigns methylation to haplotypes that WhatsHap has already resolved²⁰; MethPhaser and HapBridge learn haplotype-specific CpGs from the reads an SNV phaser has tagged, then join adjacent blocks or flag suspected switches^{21,22}; and LongHap, which co-phases SNVs, indels and SVs, adds methylation only to bridge the gaps its sequence phasing leaves²³. Because the informative CpGs are defined by the preceding sequence phase, its errors propagate into the methylation evidence, and methylation can revise a phase only after it is built. Allele-specific methylation is moreover confined to a small fraction of CpGs, varies between cells and tissues, and is read as a per-molecule probability rather than a base^{20,21}; such sparse, noisy evidence is best weighed together with the sequence alleles on the same reads.

The second challenge is accounting for errors that depend on genomic context. WhatsHap weights observations in its minimum-error-correction objective, and HapCUT2 incorporates base-quality information in its likelihood model^{7,8}. Base quality alone, however, cannot establish that a read originates from the locus to which it is aligned. Base-calling errors concentrate in homopolymers and tandem repeats, reads misplaced in segmental duplications create spurious linkages, and heterozygous calls in copy-number-altered regions can mislead the phasing graph^{24,25}. These context-dependent errors can create switches within haplotype blocks and confound compound-heterozygosity calls and haplotype-aware somatic detection^{16,19}.

A third challenge is identifying unreliable phase assignments. Statistical phasers attach posterior probabilities to phased genotypes⁵, and HapCUT2 uses posterior confidence to prune variants and split low-confidence blocks⁸. Such confidence estimates depend on the phasing model and may remain high when misplaced reads provide internally consistent evidence. Local sequence context and the surrounding evidence therefore provide additional information for assessing phase reliability. WhatsHap and Margin report no confidence, and LongPhase reports only simple read support, the number of a read’s alleles on each haplotype and a Phred-scaled tagging quality derived from them, not the reliability of each variant’s phase. Learning-based phasers, built on graph neural networks or deep reinforcement learning^{26,27}, do not close this gap. They learn, partly from simulated reads, to optimize the same fragment-cut or minimum-error-correction objective as heuristic phasers, phase SNVs only and report no confidence, and their authors note that errors from mis-alignment in repetitive regions remain unresolved²⁷.

These limitations have been underestimated, because phasers are benchmarked against Genome in a Bottle (GIAB) truth sets built largely by mapping reads to GRCh38^{28,29}, which can assess only the regions where mapping-based methods already work³⁰. The HG002 v4.2.1 release omits both sex chromosomes and 12% of the autosomes, mostly satellite arrays and segmental duplications^{29,30}, where mis-mapped reads and collapsed paralogues create false linkages^{24,25}. Its phase is also partial and partly circular: it was recorded only for the 74% of heterozygous variants on which trio-based phasing agreed with Strand-seq and HiFi phasing by WhatsHap²⁹, leaving the variants hardest for a read-based phaser the least likely to be scored. Extending benchmarks into difficult sequence has changed conclusions before. GIAB’s growth from 85% to 92% of GRCh38 exposed eight times more false negatives in a short-read call set²⁹, and the complete CHM13 reference removed tens of thousands of spurious variant calls per sample³¹. The complete diploid T2T-HG002 assembly now lifts this ceiling for phasing³⁰. It resolves both haplotypes of every chromosome, adds 701 Mb of autosomal sequence absent from v4.2.1 and phases every heterozygous site. Projected onto GRCh38³², it provides a benchmark on which phasers can be compared where they are most likely to fail.

Here we present LongPhase 2, which addresses each of these limitations. It places 5mC sites in the same weighted phasing graph as SNVs, small indels and SVs, letting methylation inform the phase while it is being built. It weights every read observation by base quality and sequence context, giving less weight to homopolymers, tandem repeats and copy-number-altered regions. It scores the phasing uncertainty of every variant, and an optional graph neural network unphases the variants it predicts to be misphased, leaving their genotypes unchanged. We evaluate it against both GIAB v4.2.1 and the T2T-HG002-derived benchmark, and it runs as a drop-in replacement for LongPhase in existing pipelines.

Results

LongPhase 2 co-phases SNVs, indels, SVs and methylation in one graph and withholds phase it predicts to be wrong

LongPhase 2 resolves haplotypes in three stages (Fig. 1). All heterozygous evidence carried by the same long reads (SNVs, small indels, SVs and allele-specific 5mC sites called by the built-in `modcall` module) enters a single phasing graph per chromosome, in which each observation is weighted by base quality and by the reliability of its sequence context, down-weighting homopolymers, tandem repeats and copy-number-altered regions (Fig. 1a,b). Each phased variant receives a phasing-uncertainty score, and a graph neural network inspects a local window around every variant whose uncertainty is high (Fig. 1c). Variants it predicts to be misphased are unphased rather than flipped, a step we call GNN correction, and a haplotype block that becomes disconnected is split; genotypes are preserved and no new phase is introduced. The outputs are phased VCFs for every variant class and haplotagged alignments (Fig. 1d).

We evaluated LongPhase 2 on HG002 nanopore R10.4.1 data down-sampled to 10–60× and on HG002 PacBio HiFi data at 10–50× (Methods). Values at 10–20× are means of ten down-sampling replicates; at 30–60× and for HiFi data and the analyses of unphased variants, one down-sampling was analysed (Methods). Phasing was scored against the GRCh38-based GIAB

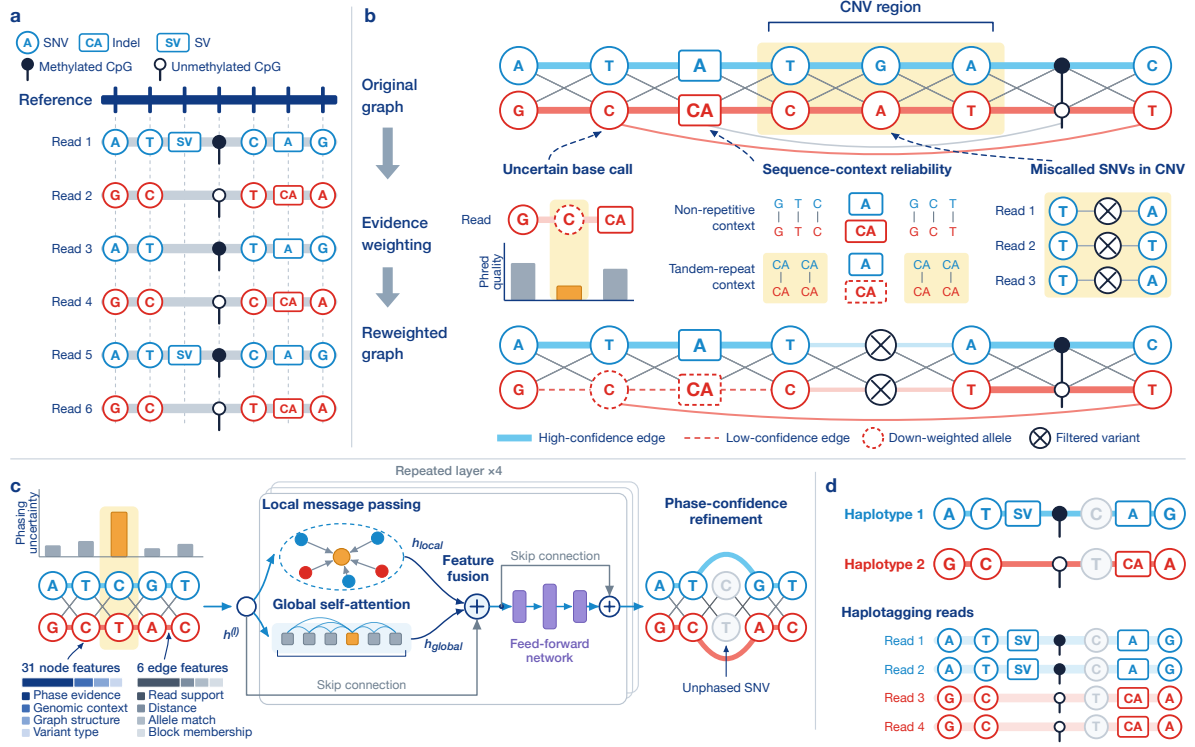


Figure 1: **Overview of LongPhase 2.** **a**, SNVs, small indels, SVs and 5mC sites carried by the same long reads enter one phasing problem; alleles are coloured by read haplotype. **b**, Weighted phasing graph. Read-supported edges link allele nodes, and each observation is weighted by base quality and sequence context; the inset shows an indel down-weighted inside a tandem repeat. **c**, Each variant receives a phasing-uncertainty score (Methods), and a graph neural network (GATv2 and Transformer layers; Supplementary Fig. 6) classifies the variants around the most uncertain ones. Variants predicted to be misphased are unphased (grey), not flipped; we call this step GNN correction. **d**, Output: phased haplotypes of all four variant classes, with the misphased variant left unphased (faded), and haplotagging reads.

v4.2.1 benchmark and against the benchmark derived from the complete diploid telomere-to-telomere (T2T) assembly of the same individual (v5.0q), which extends evaluation into segmental duplications, satellites and other regions that v4.2.1 excludes. Unless stated otherwise, results refer to v5.0q and to SNVs, the only class that every compared tool phases, and LongPhase 2 denotes the complete method, GNN correction included. We first compare LongPhase 2 with existing phasers and then analyse the contribution of its internal modules. Per-run values are in Supplementary Data 1 and Supplementary Tables 7–12.

Comparison with existing phasers

LongPhase 2 makes 2.8- to 4.8-fold fewer SNV switch errors than WhatsHap and HapCUT2. Phasing SNVs only, from identical PEPPER-Margin-DeepVariant calls, LongPhase 2 had a switch error rate of 0.082% at 10× and 0.023% at 60×, against 0.233% and 0.081% for WhatsHap 2.8 and 0.266% and 0.104% for HapCUT2 1.3.4 (replicate s.d. at 10× ≤ 0.006 percentage points), 2.8–3.7-fold below WhatsHap and 3.3–4.8-fold below HapCUT2 across 10–60× (Fig. 2b,e and Supplementary Table 8)[Required validation, not yet run. (i) Genome-wide rates include the 17 chromosomes from which training windows were drawn, on the same down-sampling replicates. Report metrics with and without GNN correction on the held-out chr17, chr21 and chr22 (switch errors, assessed pairs, retained variants, N50; longphase compare -regions) as the primary validation, and state whether these chromosomes influenced the graph constants. (ii) Matched-retention comparison: withhold the same number of variants by the network, by an entropy threshold, by a support-count threshold, at random and by HapCUT2’s per-variant confidence, and plot switch errors against retained variants and N50, separating unphasing from block splitting. (iii) Paired uncertainty (replicate-wise differences) for the fold changes at 10–20×.]. The Hamming distance ranked the tools in the same order (Fig. 2c). WhatsHap phased 0.7–1.0 percentage points more benchmark heterozygous SNVs (Fig. 2a), and the block N50 of LongPhase 2 was slightly shorter than that of the comparators at 10–20× (0.84 versus 0.95 Mb for WhatsHap at 10×) and slightly longer at 30–60× (2.86 versus 2.72 Mb at 60×; Fig. 2d). Whole-genome phasing took 3 min at 10× and 15 min at 60×, against 26 and 95 min for HapCUT2 and 26 and 115 min for WhatsHap [Runtimes have no exact source: they were read from the earlier Fig. 2e (pptx), which the regenerated Fig. 2 no longer contains, and supplementary.xlsx holds none. Supply the timing table (threads, hardware, wall-clock and peak memory per tool and coverage) as a Supplementary Table, or remove this sentence and the speed claim in the abstract. The repository README times phase of v2.0 at 24 threads as 39 s at 10× and 180 s at 60×, about five times faster than the values quoted here]. Of three further phasers compared at 10–20× (Fig. 2f and Supplementary Fig. 10), Ralph²⁷ and GCphase³³ had 2.5–3.5-fold and 21–26-fold higher switch error rates than LongPhase 2 [Hamming distances near 40% are close to random phasing; confirm that GCphase was run with the parameters its authors recommend for nanopore data before reporting it]. Margin⁹ had the lowest rate at 10× (0.069% versus 0.082%), which coincided with 3.2-fold shorter blocks (N50 0.26 versus 0.84 Mb); from 12× upwards LongPhase 2 made fewer switch errors with 2.1–2.8-fold longer blocks.

The separation between phasers depends on the benchmark. Scored against GIAB v4.2.1, the three tools converged on 1,600–2,200 switch errors from 30× upwards, a ratio of 1.2–1.3 between the highest and lowest rate; scored against v5.0q, the same VCFs differed 4.3–4.8-fold (Fig. 3a,b and Supplementary Table 9). HapCUT2 made more switch errors under v5.0q at every coverage and WhatsHap at every coverage below 60× (both 1.6-fold more at 10×), whereas LongPhase 2 made fewer (1,518 versus 1,820 at 10×; 496 versus 1,676 at 60×). Hamming distance was nearly unchanged between benchmarks (Fig. 3c,d), consistent with the additional v5.0q switch errors being isolated switches; because error locations have not been intersected, the contributions of regional coverage, variant representation and phase discordance remain unresolved[Score the regions and alleles common to both benchmarks separately from the benchmark-specific ones, and intersect switch-error locations across tools and benchmarks].

With indels, LongPhase 2 phases SNVs 3.0- to 3.7-fold more accurately than WhatsHap at similar contiguity. WhatsHap phases indels by default, which lengthened its blocks (N50 2.72 to 4.20 Mb at 60×) but raised its SNV switch error rate by 17–20% (Fig. 4a,b). Phasing the same indels, LongPhase 2 made 3.0–3.7-fold fewer SNV switch errors than WhatsHap, whereas WhatsHap phased 2.4–4.5 percentage points more benchmark heterozygous indels (Fig. 4c) and built 14–28% longer blocks. Adding SVs and 5mC sites, which WhatsHap does not phase, brought the block N50 of LongPhase 2 to within 2–19% of that of WhatsHap with indels (4.10 versus 4.20 Mb at 60×) at 3.0–3.7-fold fewer switch errors (Fig. 4b,d and Supplementary Table 10). MethPhaser²¹, which adds methylation evidence to existing SNV blocks, left the block N50 and switch errors of the LongPhase 2 SNV phasing it was given unchanged, whereas joint SNV and 5mC phasing lengthened blocks by 8% (Supplementary Fig. 11)[The MethPhaser values in MethPhaserCompare.jsx are identical, to every printed digit and at all six coverages, to the SNV-only LongPhase 2 run without GNN correction given to MethPhaser as input (Supplementary Data 1, replicate 1). Confirm that the MethPhaser output VCF was scored and that MethPhaser ran to completion (it is expected to merge blocks); if it did not, re-run it before this comparison is reported].

LongPhase 2 makes 2.5- to 3.5-fold fewer switch errors than WhatsHap on PacBio HiFi data. On SNV-only phasing of HG002 PacBio HiFi data, with the network trained on nanopore data alone, LongPhase 2 had a 2.5–3.5-fold lower switch error rate than WhatsHap (0.052% versus 0.128% at 10×; 0.042% versus 0.125% at 50×), whose rate did not decrease with coverage, and a lower Hamming distance at every coverage (Fig. 5b,c,e and Supplementary Table 11). Block N50 was within 3% of that of WhatsHap, and WhatsHap phased 0.4–0.7 percentage points more SNVs (Fig. 5a,d)[The HiFi call sets are nearly identical at every coverage (2,367,609–2,367,619 heterozygous SNVs from 10 to 50×; Supplementary Table 7), which suggests that variants were called once and the same calls phased at every down-sampled coverage, unlike the nanopore runs. Confirm, and state the HiFi data source, caller and procedure in Methods; if one call set was used, the phased fraction at 10× (87%, against 78% on nanopore) is not comparable between platforms]. HiFi blocks were 4.6 times shorter than nanopore blocks at 50× (0.55 versus 2.51 Mb), as expected from the shorter reads.

LongPhase 2 and WhatsHap leave different SNVs unphased. LongPhase 2 left more heterozygous SNV calls unphased than WhatsHap (210,683 versus 142,313 at 60×), and the two sets overlapped only partly: 88,692 were left unphased by both, 121,991 by LongPhase 2 alone and 53,621 by WhatsHap alone (Fig. 6a). Each tool’s set was enriched in genomic intervals containing switch disagreements between the other tool and the truth set, increasingly with coverage: 4.3- to 9.6-fold (10 to 60×) for SNVs left unphased by LongPhase 2 alone and 2.0- to 9.5-fold for those left unphased by WhatsHap alone (Fig. 6b). This measures regional co-location, not site-level correctness. The two sets also differed in depth (Fig. 6c and Supplementary Fig. 12): at 60×, SNVs left unphased by WhatsHap alone had a median depth of 18×, whereas 15.6% of those left unphased by LongPhase 2 alone lay at the depth cap of the variant caller, a pile-up typical of reads mismapped from paralogous sequence[the median depths and the zero-entropy share among WhatsHap-only sites (57.6%) are quoted from JHL’s commit message of 2026-09-27 and are not in any committed table; add the per-set summary to Supplementary Data 1].

Contribution of the internal modules

GNN correction removes a third of the remaining switch errors at a small cost in contiguity. Without GNN correction, the graph algorithm alone had a switch error rate of 0.121% at 10× and 0.034% at 60×, already 1.9–2.5-fold below WhatsHap and 2.2–3.3-fold below HapCUT2 (Supplementary Fig. 13 and Supplementary Table 8). GNN correction lowered it by a further 31–33% at every coverage, at a cost of 0.1–0.3 percentage points of phased SNVs and of 5–9% of block N50 at 10–20× and 2–5% above. On HiFi data it removed 19–28% of the switch errors at a cost of 0.1 percentage points of phased SNVs and 1–2% of block N50 (Supplementary Fig. 13e,f).

Methylation lengthens blocks at a similar error rate, and indels trade accuracy for contiguity. We phased seven combinations of variant classes, each with and without GNN correction (Supplementary Figs. 14 and 15 and Supplementary Table 10). Without GNN correction, adding 5mC lengthened the 60× block N50 by 8%, SVs by 1%, indels by 49% and all three by 75% (2.91 to 5.10 Mb). Only indels raised the SNV switch error rate, from 0.034% to 0.043% at 60× and from 0.121% to 0.155% at 10×[For configurations that phase indels, compare computes the Hamming distance over all common heterozygous variants, indels included (CompareProcess.cpp:579), whereas switch errors are SNV-only. Every Hamming value of an indel-containing configuration in this section, in Figs. 3c,d and 4d and in Supplementary Tables 9 and 10, therefore mixes SNV and indel phasing. Re-score these runs with SNV-only Hamming distance and report indel phasing accuracy separately]. GNN correction lowered the switch error rate of every configuration by 30–42% and reduced the increase attributable to indels by 41–67%; it also split blocks that unphased indels had bridged, shortening indel-containing configurations by 12–22%. After GNN correction, SNV+5mC phasing had a switch error rate similar to that of SNV-only phasing (497 versus 496 switch errors at 60×) with 8% longer blocks, and four-class phasing reached 0.027% with 43% longer blocks (4.10 versus 2.86 Mb). LongPhase 2 phased 30.8–45.5% of benchmark heterozygous indels, more than 99% of the 5mC sites called by modcall and 68–70% of the heterozygous SVs called by Sniffles2[The SV and 5mC phased fractions were

read from the earlier Fig. 4c,d (pptx) and have no exact source: supplementary.xlsx lists only the called SV and 5mC totals (Supplementary Table 7). Add the phased SV and 5mC counts to Supplementary Data 1 or drop these two clauses].

SNVs withheld by GNN correction are enriched outside assembly-assessable regions. GNN correction unphased 59,000–64,000 SNVs per genome. Classified against the v5.0q benchmark VCF, 81% (10×) to 93% (60×) were absent from it, 3–5% were homozygous and 14% to 4.0% (8,673 and 2,411) were heterozygous (Supplementary Fig. 16a)[The classification counts benchmark VCF records outside the benchmark BED, whereas compare scores only inside it: compare puts the loss of phased benchmark SNVs at 60× at 2,055 (2,193,366 before and 2,191,311 after GNN correction, Supplementary Data 1), and the assembly analysis below finds only 549 unphased SNVs inside the benchmark BED, of which 413 are benchmark variants. The two analyses also count different totals at 60× (60,267 here, 56,076 in the assembly analysis of the same replicate). State the matching rule and SNV filter of each and reconcile the totals]. Because v5.0q is derived from the T2T-HG002 assembly, absence means either that the assembly carries no variant or that it does not align to GRCh38 at the site. Of 56,076 SNVs unphased at 60×, 11.8% lay where both assembled haplotypes align one-to-one to GRCh38, against 92.6% of a random sample of SNVs phased before GNN correction (odds ratio 94; Supplementary Fig. 16b and Supplementary Table 12). The difference held on every autosome (Supplementary Fig. 16c), and the unaligned sites clustered in pericentromeric satellites, most densely at chr11:51–54 Mb, chr12:35–36 Mb and chr18:15–20 Mb. Of the 6,601 unphased SNVs inside the alignment, 3,421 (52%) had no assembly variant record at the same coordinate[v5.0q is derived from the same assembly; this analysis extends coverage beyond the benchmark BED; it is not independent confirmation. The dipcall BED is the raw alignment and does not exclude the assembly errors that the benchmark curation removes. Rebuild this analysis before submission from one site-level table: autosomes only (chrX and chrY are currently included); a control drawn from SNVs that stayed phased, excluding the removed ones (the current background is sampled from all SNVs phased before GNN correction, run_t2t_validation.sh:75); and an allele-aware comparison with the assembly VCF that handles overlapping complex variants (the current test matches the start coordinate only, line 104). Derive every panel of Fig. 6 from that table]. Removed benchmark heterozygotes often had weak phasing evidence, 43% (10×) to 22% (60×) with a phasing entropy of exactly 0, whereas clustering within 50 bp or low genotype quality marked 46–49% of benchmark-absent calls (Supplementary Fig. 17)[these prevalences were read from the earlier Fig. 6b,c (pptx), now Supplementary Fig. 13; replace with exact values and regenerate the figure].

Discussion

LongPhase 2 rests on two ideas: that every class of heterozygous evidence a read carries belongs in one haplotype model, provided that each observation is weighted by how far it can be trusted, and that a phaser should withhold phase where its own output is unreliable. The first extends observations made for methylation^{2,20,21} and for SVs^{11,34} that SNVs alone leave blocks stranded where heterozygosity is low; unlike LongHap, which adds methylation after sequence phasing to

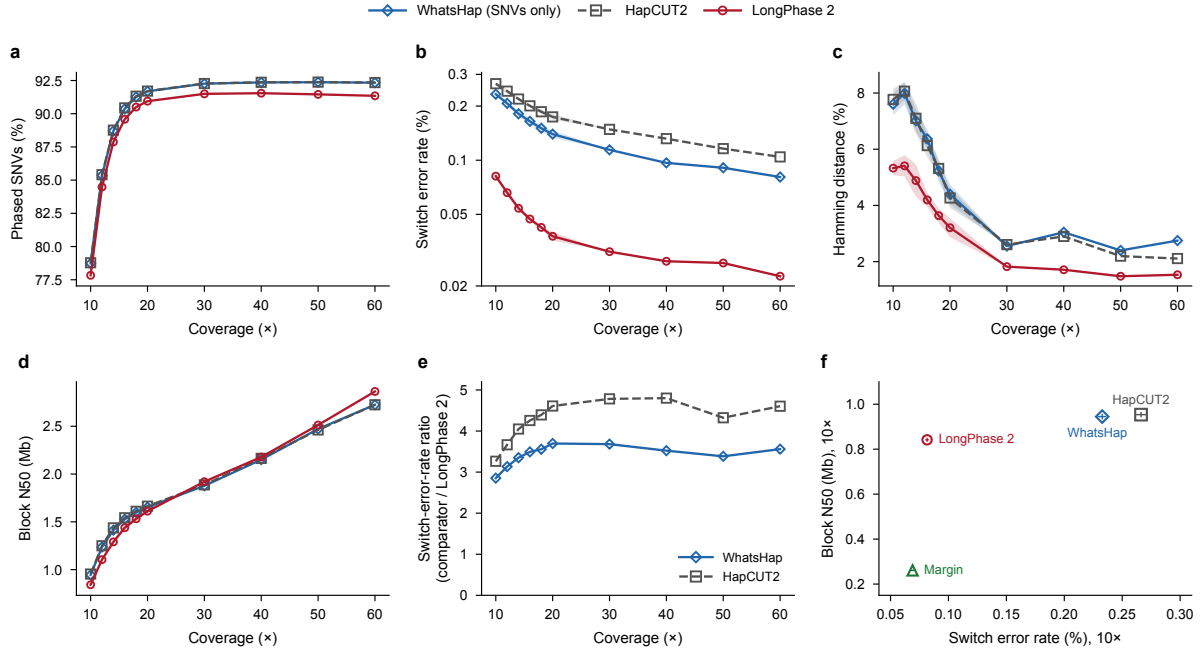


Figure 2: SNV phasing compared with WhatsHap, HapCUT2 and Margin. HG002 nanopore R10.4.1 reads down-sampled to 10–60×; SNV-only phasing of identical variant calls, scored against the T2T-HG002-derived v5.0q benchmark. Values at 10–20× are means of ten down-sampling replicates, with the s.d. as a light band (often narrower than the line); values at 30–60× come from one down-sampling of the 70× data set (Methods). Open symbols of different shape and size remain visible where tools coincide, as WhatsHap and HapCUT2 do in **a** and **d**. **a**, Fraction of benchmark heterozygous SNVs phased. **b**, Switch error rate (log scale). **c**, Block-wise Hamming distance. **d**, Phase-block N50. **e**, Switch error rate of each comparator divided by that of LongPhase 2. **f**, Switch error rate against block N50 at 10×; Ralph and GCphase in Supplementary Fig. 10.

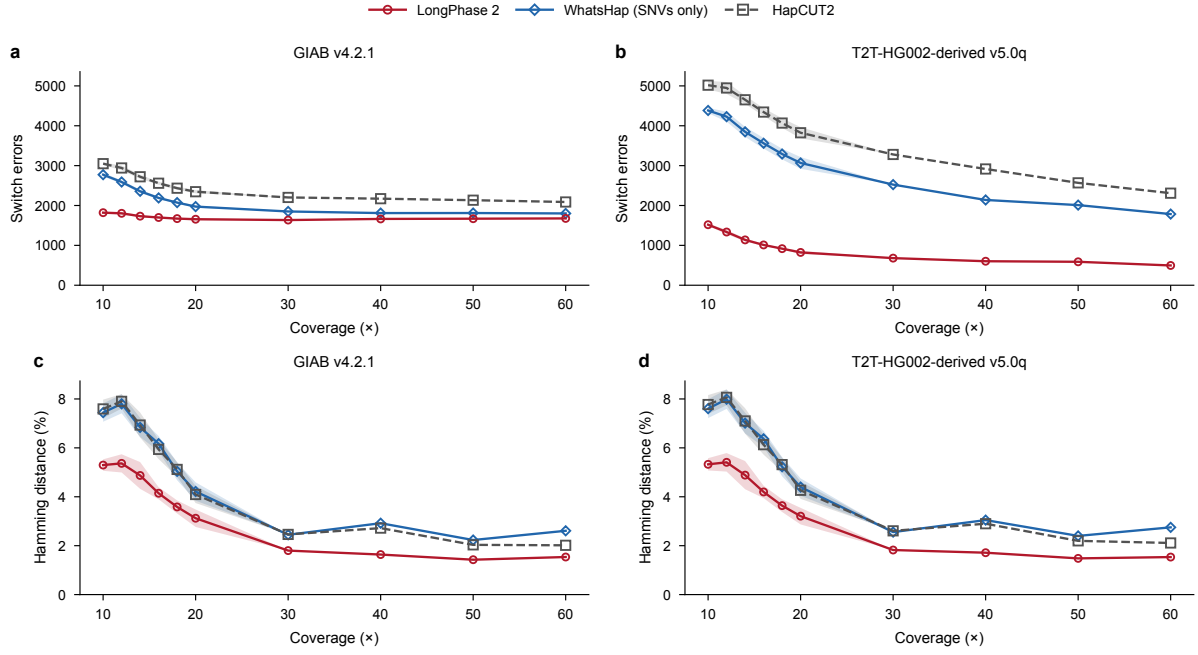


Figure 3: **The same phased VCFs scored against GIAB v4.2.1 and the T2T-HG002-derived v5.0q benchmark.** SNV-only phasing of HG002 nanopore R10.4.1 data; replicates as in Fig. 2. **a,b**, Genome-wide switch errors under v4.2.1 (**a**) and v5.0q (**b**), on a shared axis. **c,d**, Hamming distance under v4.2.1 (**c**) and v5.0q (**d**).

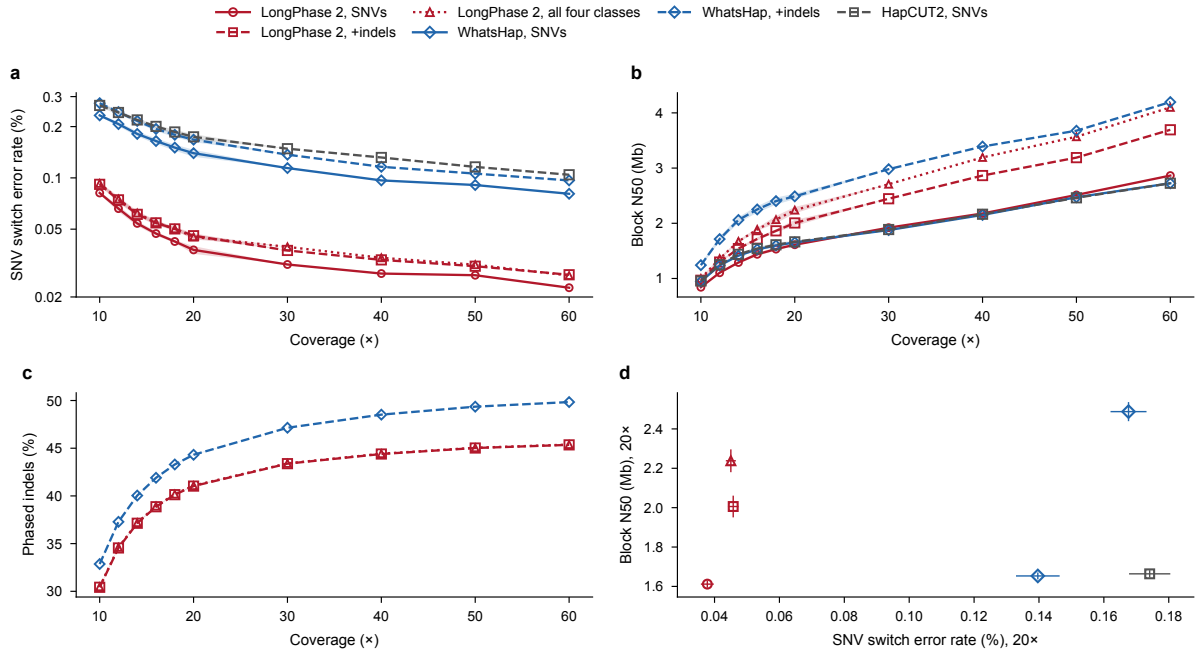


Figure 4: **Co-phasing compared with WhatsHap.** HG002 nanopore R10.4.1 data; LongPhase 2 phasing SNVs only, SNVs and indels, or all four classes (SNVs, indels, SVs and 5mC), against WhatsHap phasing SNVs only or SNVs and indels; HapCUT2 for reference. Accuracy is scored on SNVs against v5.0q; replicates as in Fig. 2. **a**, SNV switch error rate (log scale). **b**, Phase-block N50. **c**, Fraction of benchmark heterozygous indels phased. **d**, SNV switch error rate against block N50 at 20x (means of ten replicates with s.d.).

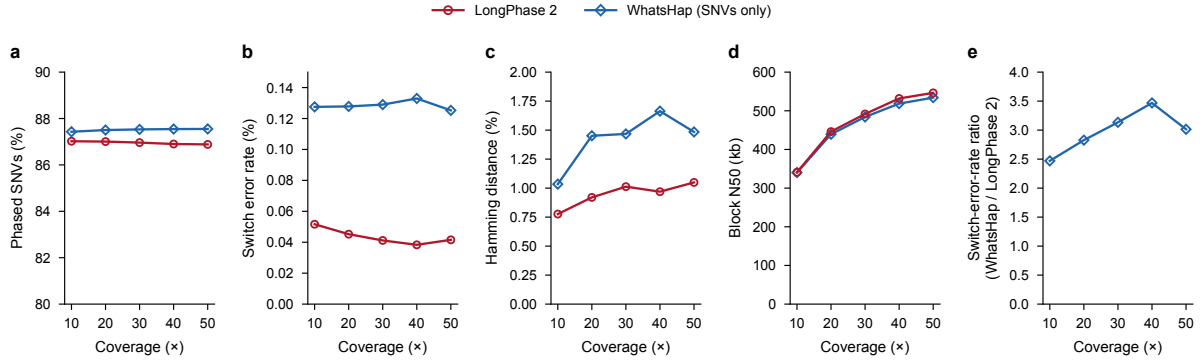


Figure 5: **Phasing of PacBio HiFi data compared with WhatsHap.** HG002 PacBio HiFi (Revio) reads down-sampled to 10–50 $\times$, one replicate per coverage[*state whether one call set was used for every coverage*]; SNV-only phasing scored against v5.0q; LongPhase 2 uses the network trained on nanopore data. Panels as in Fig. 2a–e. **a**, Fraction of benchmark heterozygous SNVs phased. **b**, Switch error rate. **c**, Block-wise Hamming distance. **d**, Phase-block N50. **e**, Switch error rate of WhatsHap divided by that of LongPhase 2.

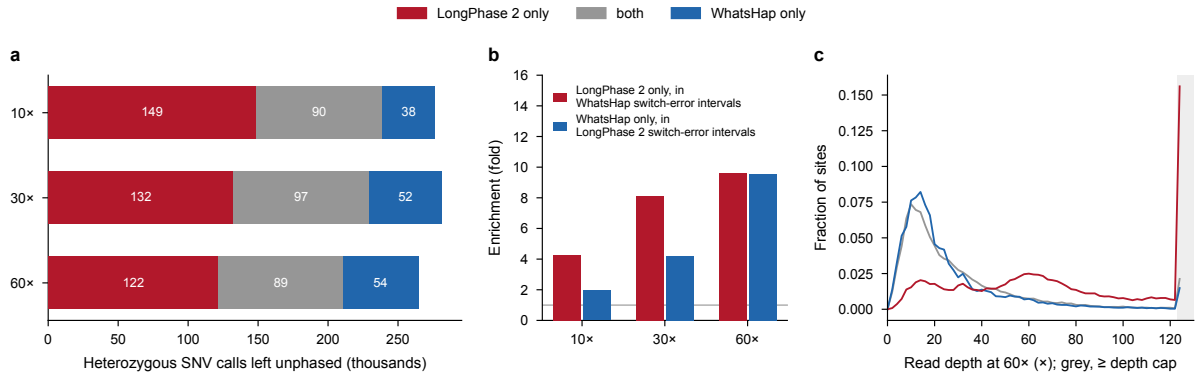


Figure 6: **SNVs left unphased by LongPhase 2 and by WhatsHap.** HG002 nanopore R10.4.1 data, SNV-only phasing, replicate 1; heterozygous SNV calls present in both output VCFs. **a**, Calls left unphased by LongPhase 2 only, by both tools or by WhatsHap only. **b**, Enrichment of each tool-specific set in genomic intervals containing switch disagreements between the other tool and the truth set, relative to that tool’s genome-wide rate; a measure of regional co-location, not of site-level correctness. **c**, Read-depth distribution of the three sets at 60 $\times$ (2 $\times$ bins); the grey bin collects sites at or above the depth cap of the variant caller.

bridge the blocks it leaves²³, LongPhase 2 weighs methylation and sequence evidence together as the phase is built. The evidence classes did not contribute alike. Methylation lengthened blocks at a similar switch error rate, whereas indels contributed most of the additional contiguity and most of the additional error, as they did for WhatsHap; this is consistent with difficulties introduced by indel evidence, although limitations that both tools share in representing or modelling indels remain possible. Joint phasing is therefore a choice between contiguity and accuracy, which GNN correction narrows and more accurate indel calling^{15,35} would narrow further.

The second idea replaces a binary phased-or-unphased state with a graded one, but only partly. LongPhase 2 writes the phasing entropy and weighted vote totals of every phased variant, in the spirit of the posteriors of HapCUT2⁸ and statistical phasers⁵, yet many removed benchmark heterozygotes had an entropy of exactly 0, unanimous but thin support that a high-entropy threshold alone would not flag. The network reads the evidence across a window rather than the likelihood of the phasing model, as HapCUT2’s pruning does, but whether it withholds better than simpler rules (entropy or support-count thresholds, random removal or HapCUT2’s confidence) at matched numbers of withheld variants has not been tested, and its phase-error score is neither calibrated nor reported. Because GNN correction only withholds phase, part of its gain follows by construction; Margin’s lower switch error rate at 10×, which coincided with three-fold shorter blocks, shows why error must be read alongside contiguity. What GNN correction withheld lay mostly outside the regions that the assembly-derived truth set can assess, where phase cannot currently be verified, and its benefit carried over to HiFi data from the same individual.

Several limitations bound these conclusions. All data come from one individual, HG002; the network was trained on its nanopore data, with labels from the evaluation benchmark and windows from the same down-sampling replicates, and the genome-wide rates include the 17 chromosomes that contributed training windows (Methods). The gains therefore require confirmation on the held-out chromosomes, on an independent individual and benchmark with frozen weights and thresholds, and above 20×. The HiFi values come from one down-sampling per coverage, and the HiFi runs appear to share one call set. Benchmark choice changed the magnitude of disagreement and the separation between methods^{29,30}; the contributions of regional coverage, variant representation and phase discordance remain unresolved, and comparisons should report both truth sets and stratify errors by genomic context²⁵. For indels, SVs and 5mC we report how many were phased, not whether they were phased correctly, and corrections to SV and 5mC phase are unsupervised. 5mC co-phasing was assessed only on the autosomes of a male lymphoblastoid cell line²¹, and the comparison with MethPhaser awaits verification. LongPhase 2 assumes a diploid genome with balanced allele fractions.

As a drop-in replacement in pipelines built on LongPhase^{14,16,18,19}, LongPhase 2 adds per-variant phasing evidence without interface changes. The most direct use is in haplotype-aware somatic calling^{16,19}, which accepts inherited phase without qualification: withholding phase at the loci where switch errors concentrate should keep spurious germline phase out of haplotagged tumour reads. Testing that, by varying only the germline phaser upstream of a fixed somatic caller, and providing a calibrated per-variant confidence that downstream tools can read, as they

316 read genotype quality, are the next steps.

317 **Methods**

318 **Overview of LongPhase 2**

319 LongPhase 2 is a diploid germline phaser that takes long-read alignments, a reference genome and
320 heterozygous variant calls, and returns phased VCFs and haplotagged alignments (Fig. 1). SNVs,
321 small indels, structural variants (SVs) and allele-specific 5-methylcytosine (5mC) observations
322 are represented together in one weighted graph per chromosome. Initial haplotypes are resolved
323 from weighted multi-neighbour votes, reads are used for a second phase-consistency pass, and a
324 graph neural network (GNN) then scores local phase errors. Variants predicted to be unreliable
325 are unphased, never flipped; if this disconnects a haplotype block, the surviving components
326 receive updated phase-set identifiers. The detailed module rules are given in Supplementary
327 Methods 1–6. The method extends the graph formulation of LongPhase 1¹¹.

328 **Input variant calls**

329 Heterozygous SNVs and small indels were taken from PEPPER-Margin-DeepVariant⁹; homozy-
330 gous and filtered records were carried through unchanged but did not enter the graph. Indels in
331 short tandem repeats were retained as evidence but down-weighted, because a misplaced repeat
332 indel can link the wrong alleles. SVs were taken from Sniffles2¹² or cuteSV¹³ with supporting
333 read names; homozygous SVs, duplicated positions and SVs coincident with SNVs were excluded.
334 The exact eligibility and repeat annotation rules are in Supplementary Method 2.

335 **Calling allele-specific methylation**

336 Modification probabilities were read from the MM/ML tags produced by Dorado³⁶ and carried
337 through alignment. A CpG was called methylated when its 5mC probability was at least 0.8,
338 unmethylated when it was at most 0.2, and ambiguous otherwise. Forward- and reverse-strand
339 observations were mapped to one CpG coordinate. A site was retained as a candidate allele-
340 specific site when methylated and unmethylated reads were sufficiently balanced and the am-
341 biguous fraction was small:

$$\frac{\min(M_R, U_R)}{\max(M_R, U_R)} \geq 0.6, \quad \frac{A_R}{M_R + U_R + A_R} \leq 0.2, \quad (1)$$

342 where M_R , U_R and A_R are the three read counts. Candidates were then required to co-segregate
343 with a neighbouring heterozygous marker. For a pair of sites, writing RR , RA , AR and AA for
344 the four observed state combinations, linkage required

$$\frac{\max(RR + AA, RA + AR)}{RR + AA + RA + AR} \geq 0.9. \quad (2)$$

345 The same criterion was used for SNV-anchored and methylation-only linking. Only CpG din-
346 ucleotides were considered. Weak candidates were admitted in a later expansion pass, and
347 consecutive CpGs were merged into one locus so that both strands shared one read list. The

resulting methylated and unmethylated read names and depths were supplied to the phasing graph. The depth bound and expansion rules are in Supplementary Method 1.

Read-level allele observations and filtering

Alignments with mapping quality below 1, unmapped, secondary or duplicate records were excluded. SNV alleles came from the aligned base and its Phred quality; indel alleles from the CIGAR; SV alleles from a matching CIGAR event or the caller’s read list; and 5mC alleles from the methylation read lists. Supplementary alignments were retained as evidence, so when two alignments of the same read overlapped by more than 20% of their combined span, the shorter alignment was discarded. Under the nanopore setting, a downstream SNV was removed when two SNVs at most 2 bp apart both lay in homopolymers of length at least three. A copy-number-aware filter identified intervals bounded by excess front- or back-clipping and removed variants whose alternate-supporting reads had an unusually high mismatch load relative to reference-supporting reads. Specifically, a read’s mismatch load R_m counted the heterozygous sites in the interval at which it carried the alternate allele, and each variant was scored by

$$\text{MR}(v) = \frac{\bar{R}_m^{\text{ALT}}}{\bar{R}_m^{\text{REF}} + \bar{R}_m^{\text{ALT}}}, \quad (3)$$

where the barred terms are the mean loads of reads supporting the ALT and REF alleles of v . Sites with $\text{MR}(v) \geq 0.7$ were removed from all reads. The three-state interval scanner and its constants are specified in Supplementary Method 2.

Weighted phasing graph

Each heterozygous variant contributed reference and alternate allele nodes. For each read, every allele was connected to the next 35 observed variants, with no edge-distance cap. Variants followed by a gap of more than 300 kb to the next heterozygous variant were skipped during haplotype resolution, casting no votes and receiving no haplotype or entropy. A read contributed weight 1 when both observations had quality at least 12 and weight 0.1 otherwise. Indel and 5mC observations and SV alternate observations were assigned fixed quality 60; reference-allele SV observations were assigned 30. Thus all such observations have full weight at the default threshold, while reference-SV evidence is down-weighted only at a higher threshold.

For a variant pair (u, v) , the four allele-pair supports were accumulated as w_{rr} , w_{ra} , w_{ar} and w_{aa} . The two possible pairings have support $P = w_{rr} + w_{aa}$ and $Q = w_{ra} + w_{ar}$, and their ambiguity is $s = \min(P, Q) / \max(P, Q)$.

Haplotype resolution by multi-neighbour voting

Variants were processed in coordinate order. Each variant cast a vote to each downstream neighbour for the pairing with larger support, except when $s > 0.7$; SNV–5mC pairs used a 0.3 threshold and required at least one read-equivalent of support. Votes had weight 1 by default, 20 for near-unanimous support ($s \leq 0.1$ with total support at least 1, or one pairing below 1 while the other was at least 1), and 0.1 when the voting variant was a tandem-repeat indel. The

vote was translated using the already assigned haplotype of the upstream variant, producing totals $h_1(v)$ and $h_2(v)$. When more than three votes had support of at most 1 read-equivalent, and at least one qualifying vote remained, only votes with $s < 0.2$, weight at least 1 and a non-indel voting variant were retained. The larger total assigned the haplotype. A tie beyond the last accepted edge opened a new block; a tie within an existing block left the variant unphased and cast no votes. The first variant position defined the phase-set identifier. The constants were fixed by a pre-study sweep on HG002 at 10–60× and were not re-tuned for the reported experiments; the calibration is described in Supplementary Method 3.

Read-based correction and phasing entropy

After the graph assignment, each read was assigned to a haplotype by a weighted majority of its phased SNV and SV alleles (weight 1), indel alleles (weight 0.1) and 5mC alleles (not used for this pass). Reads were retained only with a majority above 0.65 and total evidence above 1. For each variant, the consistency of the two read populations was

$$\rho = \frac{\max(n_1, n_2)}{n_1 + n_2},$$

where n_1 counts haplotype-1 reads carrying REF plus haplotype-2 reads carrying ALT, and n_2 counts the converse. The majority pairing assigned the phase only when $\rho > 0.75$; otherwise the variant was left unphased.

For each phased variant, phasing entropy was calculated from the weighted haplotype totals after the single-read guard:

$$\text{PE}(v) = - \sum_{i \in \{1,2\}} p_i \log_2 p_i, \quad p_i = \frac{h_i}{h_1 + h_2}. \quad (4)$$

It ranges from 0 for unanimous support to 1 bit for equal support. When $h_1 + h_2 = 0$, PE was set to 0 and the case was identified from the accompanying h_1, h_2 fields; this occurs only at the first variant of a block. PE, h_1 and h_2 were written to the VCF, and accepted voting edges with their accumulated weights formed the GNN input graph.

Graph neural network for phase-error detection

The GNN operated on merged SNV/indel, SV and 5mC records. A window of at most 41 variants (82 allele nodes) was opened around every phased variant with $\text{PE} \geq 0.8$, extending 20 variants on either side. It assigned a non-calibrated phase-error score to each variant. The 31 node features summarized phasing evidence, sequence context, graph structure and the four variant classes; six edge features summarized vote weight, genomic distance, allele matching and phase-set relations. Edges were restricted to the window, symmetrized and supplemented with mean-feature self-loops. Exact definitions and normalizations are in Supplementary Tables 1–2.

The network used four GPS graph-transformer layers³⁷: a local four-head GATv2 branch^{38,39} over graph edges and a global four-head self-attention branch⁴⁰, followed by residual normalization and a 128–256–128 feed-forward block. A two-layer classifier produced the two-class softmax output. The two allele-node scores were averaged, as were scores from overlapping

windows. Only variants no further from the trigger than half the largest trigger distance in the window were scored; the same rule defined the nodes used for training loss. Nodes outside that zone supplied context only. Exhaustive architecture and inference details are in Supplementary Method 4 and Table 3.

Training windows came from the same HG002 nanopore co-phasing runs used for evaluation at 10–20× (six coverages and ten down-sampling replicates, 60 runs). Of 722,116 windows, chr1–14 and chr18–20 supplied training (602,788), chr15–16 validation (67,880), and chr17, chr21 and chr22 testing (51,448); the chromosome sets were disjoint. No test-chromosome window entered gradient updates, batch-normalization statistics or checkpoint selection. Windows contained all four variant classes, but only benchmarked SNV and indel nodes had labels, so SV and 5mC correction was transferred from neighbouring small-variant evidence. The training and evaluation truth was the v5.0q small-variant benchmark. Training windows were limited to 10–20×, whereas results also included 30–60× and PacBio HiFi data.

Labels described local phase orientation. Within each window and phase set, the first benchmarked variant anchored the orientation; later disagreements were labelled as a single-variant flip or unphase event, or as a break at the first site of a longer run. Benchmark-absent variants and SV/5mC records were masked from the loss but retained as graph context. The released model merged flip, unphase and break into one error class. Error labels accounted for 2.96% of labelled central nodes in the training split. Correct nodes were subsampled to at most four per error node, and the resulting cross-entropy used focal weighting⁴¹ with $\gamma = 1.5$ and label smoothing 0.05. Training used haplotype-swap and coverage-scaling augmentations; the optimizer, stopping rule and environment are listed in Supplementary Table 4.

Checkpoints were selected on chr15–16 by maximizing

$$S = R \left(2 - \frac{1}{P} \right) \quad (5)$$

over 81 score thresholds from 0.01 to 0.99, subject to $R \geq 0.005$, where P and R are error-class precision and recall. The selected checkpoint was epoch 44; validation and held-out classification metrics are given in Supplementary Table 4. Deployment used a score threshold of 0.30 because the only correction action is to withhold phase, whose cost differs from the net-gain objective. [State how the deployment threshold 0.30 was chosen and add its phased-fraction/switch-error trade-off across thresholds as a supplementary panel.]

GNN correction and phase-set update

A variant with mean score at least 0.30 was unphased while preserving its genotype. Bridge protection was disabled. If removing these variants disconnected a block, the surviving connected components became separate phase sets (Supplementary Method 4).

Haplotagging

Reads were tagged from the phased SNV, indel, SV and 5mC alleles they carried. Each informative observation contributed one vote, and a read was assigned to the majority haplotype when its fraction was at least 0.60; reads whose informative variants belonged to multiple phase sets or

lacked a majority were left untagged. Tagged reads received HP, PS and phasing-quality fields. SV alleles were re-derived from the CIGAR and 5mC evidence was matched by read name. Full evidence and quality definitions are in Supplementary Method 5.

Benchmark data

The benchmark used HG002 (NA24385) Oxford Nanopore R10.4.1 reads from the ONT giab_lsk114_2022.12 release: two PromethION flow cells (PAG65784 and PAG68757), SQK-LSK114, 400 bases per second and 4-kHz sampling, basecalled with Dorado dna_r10.4.1_e8.2_400bps_sup@v4.0.0 and 5mCG_5hmCG@v2; their read N50 values were 29.2 and 29.4 kb. Reads were aligned to the GRCh38 no-alt analysis set with minimap2 v2.30. The 70.4× alignment was down-sampled to 10, 12, 14, 16, 18, 20, 30, 40, 50 and 60× with samtools v1.23.1⁴². Ten replicates were used at 10–20× and one at 30–60×; variants were called separately at every coverage and replicate. SNVs and indels came from PEPPER-Margin-DeepVariant, SVs from Sniffles2 v2.8.0, and allele-specific 5mC sites from LongPhase 2. [Give the ten down-sampling seeds and the PEPPER-Margin-DeepVariant release/model, and state that any phased caller output was discarded.]

Accuracy was scored against the GIAB v4.2.1 HG002 small-variant benchmark²⁹ and the T2T-HG002-derived v5.0q benchmark^{30,32}. Each truth set was restricted to its own benchmark BED and chr1–chr22; chrX, chrY and chrM were not scored. The v5.0q set adds 701 Mb of autosomal sequence, mainly satellite and segmental-duplication sequence, that v4.2.1 does not assess³⁰.

Comparison with other methods

WhatsHap v2.8⁷ and HapCUT2 v1.3.4⁸ were run on the same alignments and variant calls; cross-tool SNV comparisons used SNV-only phasing, with indel phasing disabled in WhatsHap. LongPhase 1.0¹¹ and LongPhase 2 were run with the nanopore setting, and GNN correction used PE trigger 0.8, score threshold 0.30, no bridge protection and block splitting. Co-phasing experiments separately evaluated the seven variant-class combinations listed in Supplementary Table 10, with and without correction. Indel co-phasing was also compared with WhatsHap’s default indel phasing. Margin v2.3.1⁹, Ralph²⁷, GCphase³³ and MethPhaser²¹ were evaluated as described in Supplementary Table 5. HG002 PacBio HiFi (Revio) reads were down-sampled to 10, 20, 30, 40 and 50× and phased SNV-only with LongPhase 2, with and without correction, and WhatsHap; the GNN weights were trained on nanopore data. [Give complete comparison commands and versions, state whether HapCUT2 default likelihood pruning was enabled, whether all tools used the same machine, and whether LongPhase 1.0 was re-run for every low-coverage replicate; document the HiFi source, alignment, caller, mode and call-set handling.]

Evaluation metrics

Metrics (Supplementary Fig. 8) were computed by intersecting heterozygous query and truth records by position and REF/ALT alleles, then partitioning the intersection into blocks phased in both files; singleton blocks were ignored. Within each block, switch errors were the Ham-

ming distance between truth and query switch-encoded haplotype strings, and the switch-error rate divided their number by the assessed consecutive SNV pairs. Isolated switches and two-consecutive-switch flips were reported separately. Hamming distance was the block-wise minimum over the two orientations, divided by the number of compared variants; unlike switch rate, it included indels when present in both files. Phased fraction was the number of truth heterozygous SNVs or indels that were phased by the query divided by all truth heterozygous variants of that class. Block N50 was the span-weighted median of query phase-set spans of at least two variants, pooled over chromosomes. Switch-error positions were exported for the unphased-SNV analysis. The exact block and denominator definitions are in Supplementary Method 6. [State the exact region-filter commands and add a metric table defining truth eligibility, numerator, denominator and block handling; re-score indel-containing configurations with an SNV-only Hamming distance; define the depth, informativeness and genotype-quality bins used in Supplementary Fig. 17.]

Analysis of unphased SNVs

An SNV was counted as unphased when it was phased in the pre-correction output and unphased after GNN correction. Such sites were classified against v5.0q as absent, homozygous or heterozygous (Supplementary Fig. 16a). At 60 $\times$, they were also classified against the T2T-HG002 assembly projected to GRCh38 as outside the 1:1 assembly BED, inside without an assembly variant at the same start coordinate, or inside with one. A random sample of 50,000 SNVs phased before correction provided the background; enrichment was summarized by chromosome and 1-Mb bins. To compare tools, shared heterozygous SNVs were partitioned into sites unphased by LongPhase 2 only, WhatsHap only or both, and the fraction of each tool-specific set within the other tool's switch-error intervals was compared with the fraction of that other tool's assessed SNVs within those intervals. [Define switch-error intervals and deposit the scripts for Fig. 6 and Supplementary Fig. 16a.]

Runtime and memory

Wall-clock time and peak resident memory were measured with `/usr/bin/time -v`; reported LongPhase 2 times included alignment reading, graph construction, phasing, GNN correction and VCF writing, and excluded alignment, variant calling and down-sampling. [Record the machine, compiler, thread count, timing repeats and summary statistic, and report memory with the runtime results.]

Statistics and reproducibility

Ten nanopore down-sampling replicates at 10–20 $\times$ were summarized by their mean and standard deviation, while 30–60 $\times$ and HiFi used one down-sampling per coverage. Subsamples for each platform were drawn from the same source sequencing data set of HG002. The reported standard deviations therefore describe variation due to read subsampling, conditional on that data set, rather than variation among independent sequencing experiments. Subsampling variability was not estimated at 30–60 $\times$ or for HiFi. Tools compared at each coverage and replicate used the

same sampled reads. No statistical tests, sample-size calculation, randomization or blinding were used. GNN inference used fixed model weights and no sampling. [State the HiFi source depth and whether further replicates were possible, and report paired replicate-wise correction effects where applicable.]

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The GRCh38 no-alt reference is available from NCBI as GCA_000001405.15_GRCh38_no_alt_analysis_set.fna.gz. The GIAB HG002 v4.2.1 files are HG002_GRCh38_1_22_v4.2.1_benchmark.vcf.gz and HG002_GRCh38_1_22_v4.2.1_benchmark_noinconsistent.bed. The v5.0q release contains GRCh38_HG2-T2TQ100-V1.1_smvar.vcf.gz and GRCh38_HG2-T2TQ100-V1.1_smvar.benchmark.bed; the scored VCF is its chr1–chr22 subset, HG002_GRCh38_v5.0q_smvar.chr1_22.vcf. HG002 nanopore reads and alignments are available from ONT's giab_lsk114_2022.12 open-data release under CC BY-NC 4.0. The source URLs are listed here for reproducibility:

- NCBI GRCh38 no-alt analysis set: https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/000/001/405/GCA_000001405.15_GRCh38/seqs_for_alignment_pipelines.ucsc_ids/;
- GIAB v4.2.1: https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/release/AshkenazimTrio/HG002_NA24385_son/NISTv4.2.1/GRCh38/;
- GIAB v5.0q: https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/data/AshkenazimTrio/analysis/NIST_HG002_DraftBenchmark_defrabbV0.020-20250117/;
- ONT reads: [s3://ont-open-data/giab_lsk114_2022.12/](https://s3.amazonaws.com/ont-open-data/giab_lsk114_2022.12/).

[State the command used to derive the v5.0q chr1–chr22 subset and provide source data for Figs. 2–6 and every Supplementary figure with plotted values, including any access restrictions.]

Code availability

LongPhase 2 is open source under the GNU General Public License v3.0 at <https://github.com/twolinin/longphase>; the repository includes a demonstration data set and workflow. [Give the exact release tag and full commit hash used for the reported results and cite the archived release.] The training chain, released model, plotting code and validation scripts are not yet deposited with the public repository. [Deposit the training and figure-generation files under an OSI-approved licence, parameterize development-machine paths, and cite the deposited archive here.]

References

- [1] Eisfeldt, J., Ek, M., Nordenskjöld, M. & Lindstrand, A. Toward clinical long-read genome sequencing for rare diseases. *Nature Genetics* **57**, 1334–1343 (2025).
- [2] Akbari, V. *et al.* Parent-of-origin detection and chromosome-scale haplotyping using long-read DNA methylation sequencing and Strand-seq. *Cell Genomics* **3**, 100233 (2023).
- [3] Logsdon, G. A., Vollger, M. R. & Eichler, E. E. Long-read human genome sequencing and its applications. *Nature Reviews Genetics* **21**, 597–614 (2020).
- [4] Kolmogorov, M. *et al.* Scalable Nanopore sequencing of human genomes provides a comprehensive view of haplotype-resolved variation and methylation. *Nature Methods* **20**, 1483–1492 (2023).
- [5] Hofmeister, R. J., Ribeiro, D. M., Rubinacci, S. & Delaneau, O. Accurate rare variant phasing of whole-genome and whole-exome sequencing data in the uk biobank. *Nature Genetics* **55**, 1243–1249 (2023).
- [6] Porubsky, D. *et al.* Fully phased human genome assembly without parental data using single-cell strand sequencing and long reads. *Nature Biotechnology* **39**, 302–308 (2021).
- [7] Patterson, M. *et al.* WhatsHap: weighted haplotype assembly for future-generation sequencing reads. *Journal of Computational Biology* **22**, 498–509 (2015).
- [8] Edge, P., Bafna, V. & Bansal, V. HapCUT2: robust and accurate haplotype assembly for diverse sequencing technologies. *Genome Research* **27**, 801–812 (2017).
- [9] Shafin, K. *et al.* Haplotype-aware variant calling with PEPPER-Margin-DeepVariant enables high accuracy in nanopore long-reads. *Nature Methods* **18**, 1322–1332 (2021).
- [10] Martin, M. *et al.* WhatsHap: fast and accurate read-based phasing. *bioRxiv* (2016).
- [11] Lin, J.-H., Chen, L.-C., Yu, S.-C. & Huang, Y.-T. LongPhase: an ultra-fast chromosome-scale phasing algorithm for small and large variants. *Bioinformatics* **38**, 1816–1822 (2022).
- [12] Smolka, M. *et al.* Detection of mosaic and population-level structural variants with Sniffles2. *Nature Biotechnology* **42**, 1571–1580 (2024).
- [13] Jiang, T. *et al.* Long-read-based human genomic structural variation detection with cuteSV. *Genome Biology* **21** (2020).
- [14] Zheng, Z. *et al.* Symphonizing pileup and full-alignment for deep learning-based long-read variant calling. *Nature Computational Science* **2**, 797–803 (2022).
- [15] Zheng, Z. *et al.* Accelerated long-read variant calling with Clair3 for whole-genome sequencing. *Bioinformatics* **42** (2026).
- [16] Zheng, Z. *et al.* ClairS: a deep-learning method for long-read tumor–normal pair somatic small variant calling. *Nature Methods* **23**, 1507–1520 (2026).

- [17] Chen, L. *et al.* ClairS-TO: a deep-learning method for long-read tumor-only somatic small variant calling. *Nature Communications* **16** (2025).
- [18] Oxford Nanopore Technologies. epi2me-labs/wf-human-variation: Nextflow workflow for the analysis of human genetic variation from Oxford Nanopore sequencing data. <https://github.com/epi2me-labs/wf-human-variation> (2025). Version 2.x; phasing performed with LongPhase (`-use_longphase`).
- [19] Keskus, A. G. *et al.* Severus detects somatic structural variation and complex rearrangements in cancer genomes using long-read sequencing. *Nature Biotechnology* **44**, 247–257 (2025).
- [20] Akbari, V. *et al.* Megabase-scale methylation phasing using nanopore long reads and NanoMethPhase. *Genome Biology* **22** (2021).
- [21] Fu, Y. *et al.* MethPhaser: methylation-based long-read haplotype phasing of human genomes. *Nature Communications* **15** (2024).
- [22] Chen, H., Nie, F., Luo, F. & Wang, J. HapBridge: a methylation-guided approach for correcting switch errors and bridging phased blocks in long-read phasing. *bioRxiv* (2025).
- [23] Pfennig, A. & Akey, J. M. Methylation-aware long-read phasing significantly improves genome-wide haplotype reconstruction. *bioRxiv* (2026).
- [24] Olson, N. D. *et al.* Variant calling and benchmarking in an era of complete human genome sequences. *Nature Reviews Genetics* **24**, 464–483 (2023).
- [25] Dwarshuis, N. *et al.* The GIAB genomic stratifications resource for human reference genomes. *Nature Communications* **15** (2024).
- [26] Xue, H., Rajan, V. & Lin, Y. Graph coloring via neural networks for haplotype assembly and viral quasispecies reconstruction. In *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35 (2022).
- [27] Battistella, E., Maheshwari, A., Ekim, B., Berger, B. & Popic, V. Graph-based deep reinforcement learning for haplotype assembly with Ralphi. *Genome Research* **35**, 2617–2625 (2025).
- [28] Zook, J. M. *et al.* An open resource for accurately benchmarking small variant and reference calls. *Nature Biotechnology* **37**, 561–566 (2019).
- [29] Wagner, J. *et al.* Benchmarking challenging small variants with linked and long reads. *Cell Genomics* **2**, 100128 (2022).
- [30] Hansen, N. F. *et al.* A complete diploid human genome benchmark for personalized genomics. *Cell* **189**, 4857–4875.e31 (2026).
- [31] Aganezov, S. *et al.* A complete reference genome improves analysis of human genetic variation. *Science* **376** (2022).

- [32] Genome in a Bottle Consortium. HG002 T2T-HG002-derived small-variant benchmark (GIAB v5 draft, “v5.0q”). https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/release/AshkenazimTrio/HG002_NA24385_son/T2T-Q100/ (2025). Draft small-variant benchmark projected onto GRCh38 from the T2T-HG002 v1.1 assembly.
- [33] Luo, J., Wang, J., Zhai, H. & Wang, J. GCphase: an SNP phasing method using a graph partition and error correction algorithm. *BMC Bioinformatics* **25** (2024).
- [34] Holt, J. M. *et al.* HiPhase: jointly phasing small, structural, and tandem repeat variants from HiFi sequencing. *Bioinformatics* **40** (2024).
- [35] Kolesnikov, A. *et al.* Local read haplotagging enables accurate long-read small variant calling. *Nature Communications* **15** (2024).
- [36] Oxford Nanopore Technologies. Dorado: high-performance, easy-to-use, open source base-caller for Oxford Nanopore reads. <https://github.com/nanoporetech/dorado> (2025).
- [37] Rampášek, L. *et al.* Recipe for a general, powerful, scalable graph transformer. In *Advances in Neural Information Processing Systems*, vol. 35 (2022).
- [38] Veličković, P. *et al.* Graph attention networks. In *International Conference on Learning Representations (ICLR)* (2018).
- [39] Brody, S., Alon, U. & Yahav, E. How attentive are graph attention networks? In *International Conference on Learning Representations (ICLR)* (2022).
- [40] Vaswani, A. *et al.* Attention is all you need. In *Advances in Neural Information Processing Systems*, vol. 30 (2017).
- [41] Lin, T.-Y., Goyal, P., Girshick, R., He, K. & Dollár, P. Focal loss for dense object detection. In *IEEE International Conference on Computer Vision (ICCV)* (2017).
- [42] Danecek, P. *et al.* Twelve years of SAMtools and BCFtools. *GigaScience* **10** (2021).